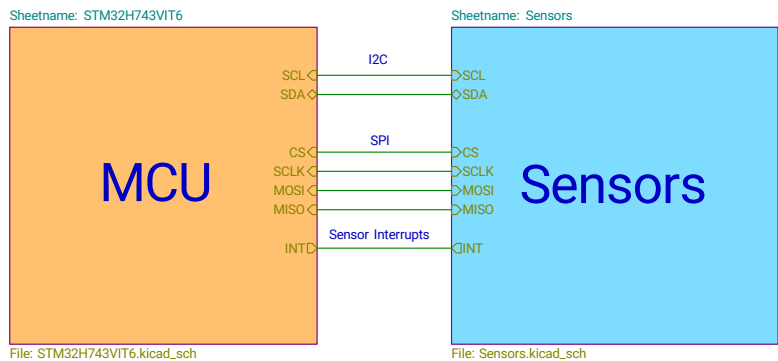
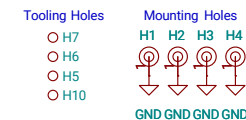
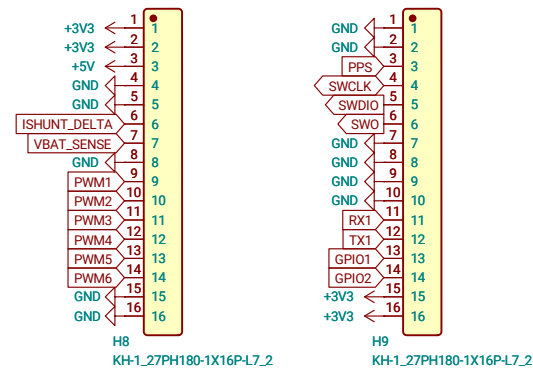




Mounting and Manufacturing Hardware

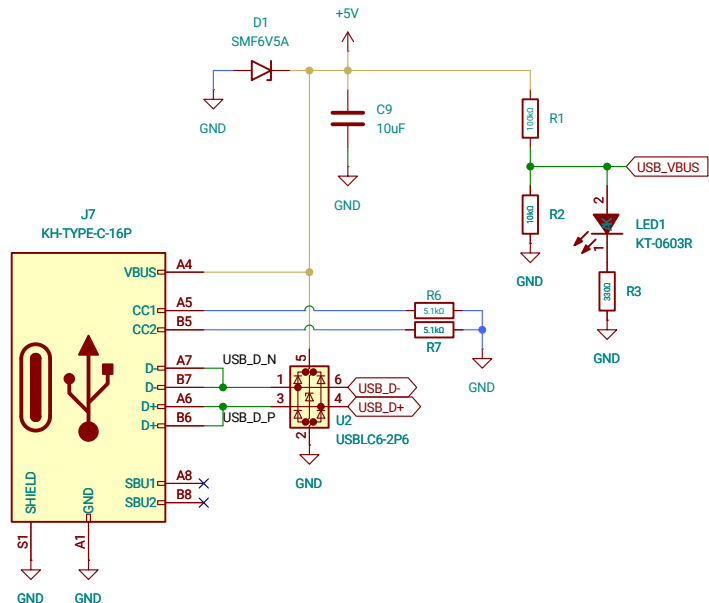


Board-to-Board Connectors - 1.27mm 16P



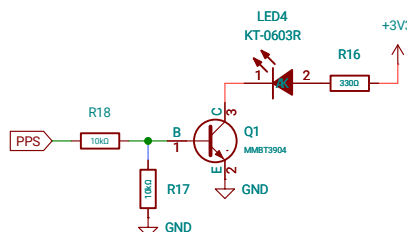
USB-C Data and 5V

Note: Route D+ and D- differentially with 90 Ohm +10%

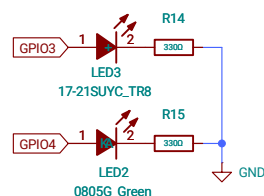


GPS Connection LED

Note: GPS can't provide sufficient current so transistor required, blinks at 1hz on GPS Lock

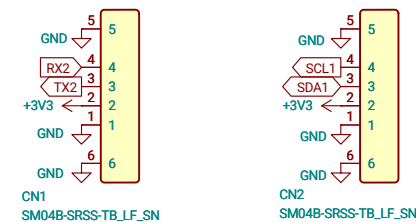


State LED



JST SH1.0 4P

Note: Breakout boards for additional functionality UART for potential LoRa telemetry



James Rhodes

Sheet: /

File: MCU Board 0.1.kicad_sch

Title: Flight Computer V6 - Sensor Board

Size: A4

Date: 2026-04-23

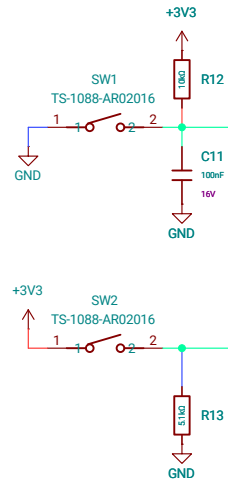
KiCad E.D.A. 10.0.1

Rev: 1

Id: 1/3

Boot and Reset Buttons

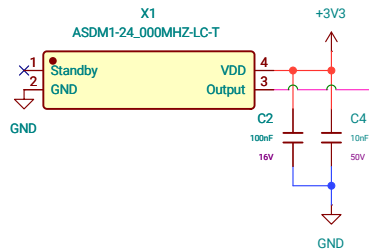
Layout: Place at board edge, easily accessible during use



Crystal Oscillator - 24Mhz

Note: Is an active oscillator

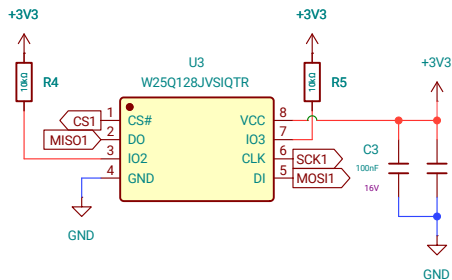
Layout: Close as possible, with GND cleared around and no traces nearby



Flash Memory - 16MB

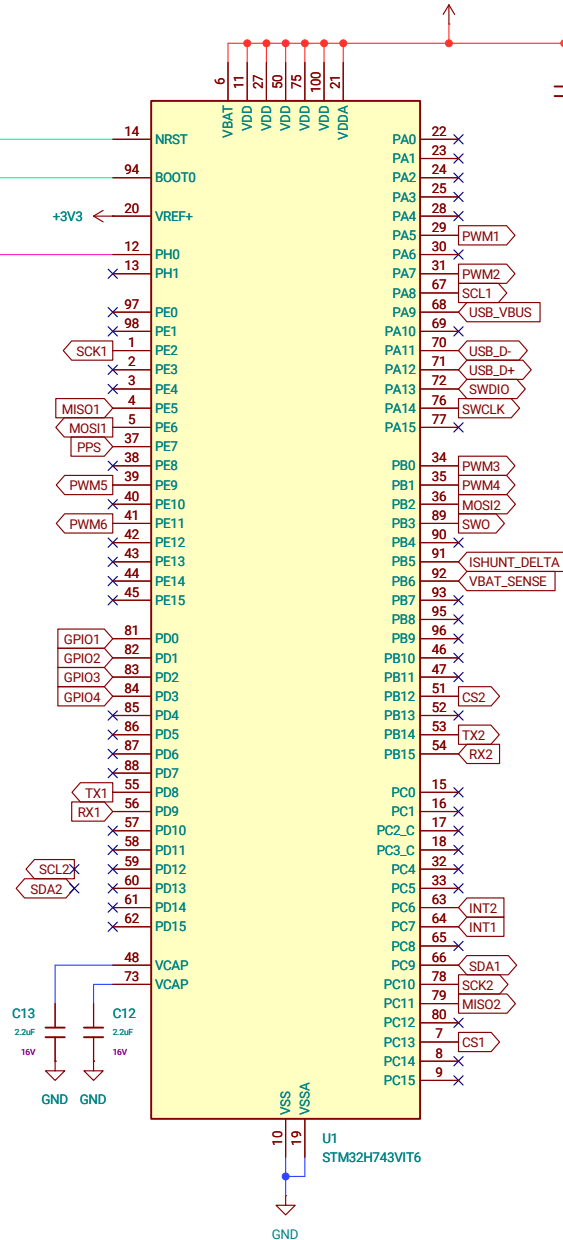
Note: SPI

Layout: Close to MCU, length match traces



MCU - STM32H743

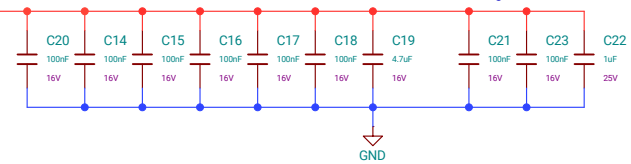
Layout: Place decoupling cap at each VDD pin



MCU Decoupling

Digital - VDD, VBAT

Analog - VDDA, VREF+



Sheet: /STM32H743VIT6/
File: STM32H743VIT6.kicad_sch

Title: MCU Components

Size: A4

Date: 2026-04-23

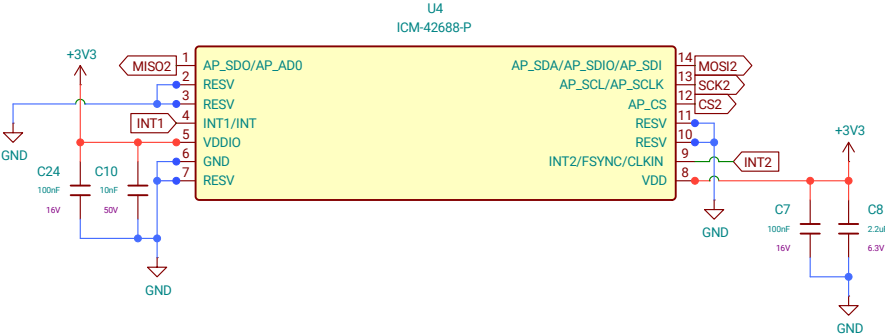
KiCad E.D.A. 10.0.1

Rev: 1

Id: 2/3

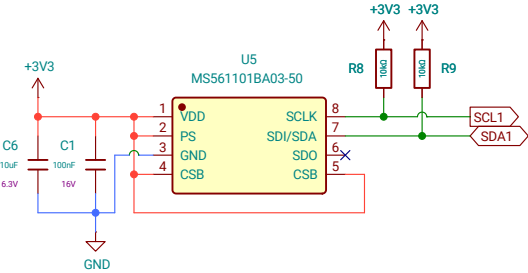
IMU - Gyro and Accelerometer +-16g

Note: Using SPI Communication Protocol
Software: Set to full accelerometer range, or implement variable ranges dependent on flight conditions



Barometer

Note: Sensitive component, keep at "ambient" pressure



Sheet: /Sensors/
File: Sensors.kicad_sch

Title:

Size: A4

Date:

Rev:

KiCad E.D.A. 10.0.1

Id: 3/3